



ACIDIC AND ALKALI SOIL — FULL MIND MAP

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1. Usar Soil — pH, Deficiencies & Treatment

- Usar soil ka pH 7 se zyada hota hai — **Bacteria, Nitrogen, Zinc ki severe kami** hoti hai. Treatment: **Pyrites, Gypsum, green manure** se hota hai (Lime NAHI, Lime acidic soil ke liye use hota hai). Tolerant paddy varieties (**Narendra Shankar Usar Paddy-1, -2**) iss soil mein grow ki jaa sakti hain. [Q1]

 **EXAM TRAP** — Lime — acidic soil ke liye use hota hai; Usar (alkaline) soil ke liye Gypsum/Pyrites use hote hain

2. Soil pH Scale — 6-7 Ideal for Crops

- Soil pH 7 se kam — **acidic**, 7 se zyada — **alkaline** hoti hai. **Fertile soil (common crops ke liye)** ka pH 6-7 **hota hai** — is range mein nutrients easily dissolve hote hain. [Q2]

3. Lime — Treatment for Acidic Soil

- Acidic soil ko cultivable banane ke liye **Lime (chuna)** use hota hai — Gypsum, Calcium, Vegetable compost se NAHI. [Q3]

4. Gypsum — Long-Term Salinity/Alkalinity Solution

- Soil salinity aur alkalinity ka solution **Gypsum use karna** hai — ye **dono short-term aur long-term solution** hai (alkalinity change karta hai, salinity remove karta hai, ~1 foot water leaching ke liye chahiye). [Q4, Q5]

5. Maximum Alkali Area — Uttar Pradesh

- India mein **maximum alkali area** **Uttar Pradesh** mein hai — Gujarat, Haryana, Punjab se zyada. [Q6]

6. Maximum Saline Soil Area — Gujarat

- India mein **sabse zyada saline/salt-affected soil area** **Gujarat** mein hai — UP, Haryana, Rajasthan, TN, AP se zyada (Desertification aur Land Degradation Atlas 2021 ke mutabik). [Q7, Q8]

7. Salinization Process — Effects on Soil

- Salinization tab hoti hai jab irrigation water evaporate hoke salts-minerals chhod jaata hai — iska effect: **kuch soils ko impermeable bana deta hai** — crop production ko badhaata NAHI, na hi groundwater level raise karta hai. [Q9]

8. Saline Soil Unsuitability (Assertion-Reason)

- India mein saline soils crops ki cultivation ke liye appropriate NAHI hain (Assertion sahi) kyunki **saline soils mein salt ka high proportion hota hai, mainly dry climate aur poor drainage ki wajah se** (Reason sahi explanation hai) — Sodium/Potassium excess, high Magnesium ratio, lime ki kami bhi hoti hai. [Q10]

9. Tea Plantation — Slightly Acidic Soil

- Tea plantation ke liye **slightly acidic, calcium-rahit soil sabse suited** hai — Alkaline, Alluvial, Regur se nahi. Tea **stagnant water ko tolerate nahi karti**, isliye hill slopes par ugaayi jaati hai jahan drainage achha ho. [Q11]